

Thomas Falconer

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Education

Technical University of Denmark

Ph.D. in Machine Learning and Economics

June 2022 – June 2025

Kgs. Lyngby, Denmark

University College London

M.Sc. in Data Science (Distinction, Top in Year)

Sept 2019 – Nov 2020

London, United Kingdom

University of Amsterdam

B.Sc. in Operations Research (Erasmus Exchange)

Sept 2016 – June 2017

Amsterdam, Netherlands

Heriot-Watt University

B.Eng. in Engineering (First Class, Top in Year)

Sept 2014 – June 2019

Edinburgh, United Kingdom

Experience

Quantitative Researcher (Algorithmic Trading)

In Commodities

June 2025 – present

Aarhus, Denmark

- Develop probabilistic ML models to forecast short-term power price movements in Europe.
- Generate signals and execution policies for continuous trading, navigating LOB dynamics, liquidity, and forecast updates.
- Apply stochastic programming over forecast distributions to optimize bidding in day-ahead and intraday auctions.
- Conduct rigorous backtesting and statistical validation to assess model reliability and strategy performance.
- Build real-time dashboards to monitor algorithms, data quality, and model drift.
- Implement CI/CD pipelines for automated model retraining, validation, and production deployment.

Machine Learning Engineer (Contract)

Freelance (During Ph.D.)

June 2022 – June 2022

Copenhagen, Denmark

- Designed and deployed end-to-end ML solutions for clients, including data ingestion, feature engineering, model development, and performance monitoring.
- Developed maintainable codebases with testing, packaging, and documentation.

Data Scientist (Algorithmic Trading)

Arenko

Nov 2020 – June 2022

London, United Kingdom

- Built and maintained flexibility trading algorithms for renewable and battery assets, using ML and constrained optimization to maximize revenue across energy markets and ancillary services.
- Implemented ensembles of forecasting models, including gaussian processes, neural networks and gradient boosting, plus simpler statistical models.
- Built MLOps pipelines in AWS, including feature store, model versioning (MLflow), low-latency model serving (FastAPI), data engineering and orchestration (Prefect) and ingestion (RDBMS) in a microservice framework.
- Created interactive visualisations of market opportunities (matplotlib, Plotly, Dash, Streamlit).
- Mentored junior data scientists and helped grow the data science team.

Machine Learning Researcher (Intern)

Invenia Labs

Apr 2019 – Sept 2019

London, United Kingdom

- Developed ML pipelines to forecast nodal price spreads for virtual/convergence bidding across US ISO markets.
- Applied graph neural networks to model grid topology for congestion forecasting and optimal power flow.
- Co-authored NeurIPS workshop paper and IEEE journal publication on ML for power systems.

Technical Skills

Analysis: pandas, Polars, NumPy; experiment design, statistical testing, feature engineering, model explainability

Modelling: scikit-learn, LightGBM, PyTorch, TensorFlow; Bayesian inference, uncertainty quantification, validation

Optimization: Pyomo, CVXPY, Gurobi; stochastic programming, constrained optimization, DRO, metaheuristics

MLOps: Docker, FastAPI, MLflow, Prefect, Argo; ETL, versioning, model serving, orchestration, monitoring

Infrastructure: AWS (S3, EC2, Lambda), PostgreSQL, CI/CD, logging (Humio), package distribution (Gemfury)

Visualization: matplotlib, Plotly, Dash, Streamlit